

ANDREW M. ROGERS

Vancouver, BC | 514-663-5690 | andrew.rogers482@gmail.com | [LinkedIn](#) | Portfolio: andrwrgers.github.io

Robotics Software Engineer (MAsc) | Embedded Systems & Production

Robotics software engineer building reliable embedded systems for products that operate outside the lab. I combine modern C++ and Python with controls, state estimation, simulation, and cross-functional product delivery.

- **Technical Core:** Modern C++ (14/17/20), Python, embedded C, real-time systems, and production firmware.
- **Robotics:** Controls, state estimation, simulation, computer vision, and motion planning across physical and simulated systems.
- **Delivery:** Experience moving robotics software from prototypes through integration, validation, release, and ongoing product support.

GlüxKind | Vancouver, BC

Software Engineer: Robotics II (Promoted within 8 months) | **Jan 2024 – Present**

- **Productization & Embedded Controls:** Helped take Rosa, GlüxKind's smart-stroller platform, from prototype-stage development to commercial production, shipping and supporting C++ firmware for motion assistance, automatic braking, and parking.
- **State Estimation & Sensor Fusion:** Build real-time estimation pipelines that turn noisy sensor data into stable signals for product control and diagnostics.
- **Simulation & Autonomy:** Build high-fidelity simulation environments to develop and validate autonomous behaviours presented in public technology demonstrations.
- **Connected Systems & Tooling:** Develop embedded support for software updates and remote diagnostics, plus Python analysis tools used to debug and validate robotics software.
- **Technical Leadership:** Lead design reviews, cross-functional integration, documentation, and production-readiness work across firmware, hardware, and product teams.

University of Toronto | Toronto, ON

Graduate Research Assistant | **May 2021 – August 2023**

- **Swarm Robotics Research:** Developed a novel, scalable exploration and mapping strategy for 40+ mobile robots, validated through custom Python simulations.
- **Hardware Implementation:** Architected and programmed millimeter-scale robots (mROBerTO) in embedded C to physically validate swarm algorithms and ensure system reliability.
- **Academic Impact:** Published findings in high-impact journals including IEEE Transactions on Robotics (TRO) and MDPI Robotics.

Safran Landing Systems | Toronto, ON

Research and Technology Engineering Intern | Aug 2019 – Aug 2020

- Developed a prototype computer vision-based inspection system for Boeing 747 nose landing gear, optimizing assembly verification times.

EDUCATION

MASc Mechanical Engineering | University of Toronto | **2023**

- **Thesis:** Occupancy Grid Mapping via Resource-Constrained Robotic Swarms: A Collaborative Exploration Strategy.

BASc Mechanical Engineering (Honours) | University of Toronto | **2021**

- **Capstone:** Engineered a 600V inverter and motor controller for Formula SAE utilizing Altium Designer and SolidWorks.

TECHNICAL SKILLS

- **Languages & Frameworks:** C++ (14/17/20), Python, MATLAB, Embedded C, OpenCV
- **Firmware/IoT:** ESP-IDF, FreeRTOS, AWS IoT, MQTT, Protobuf, BLE, I2C/UART/SPI
- **Robotics & Controls:** ROS2, Sensor Fusion (EKF, IMM Kalman), Impedance Control, Matrix Lie Groups, State Estimation (SE(3)/SO(3))
- **Data Visualization:** Dash, Plotly, Pandas, NumPy, SciPy, Folium
- **Simulation:** Nvidia Isaac Sim, Gazebo, Custom Python Environments
- **Design & Hardware:** SolidWorks, CATIA V5, Altium Designer, I2C/UART/RS232, JTAG, Additive Manufacturing (FDM/SLA).
- **Tools:** Git/GitHub (CI/CD), Linux, Linear, Notion, Jira

PUBLICATIONS

- **A. Rogers**, K. Eshaghi, G. Nejat, and B. Benhabib, "Occupancy Grid Mapping via Resource-Constrained Robotic Swarms: A Collaborative Exploration Strategy," MDPI Robotics, vol. 12, no. 3, Art. no. 3, Jun. 2023, doi: 10.3390/robotics12030070.
- K. Eshaghi, **A. Rogers**, G. Nejat, and B. Benhabib, "Closed-Loop Motion Control of Robotic Swarms – A Tether-Based Strategy," IEEE Transactions on Robotics, vol. 38, no. 6, pp. 3564–3581, Dec. 2022, doi: 10.1109/TRO.2022.3181055.